Final Report SDS336

Problem Definition

Determining a teacher’s salary is a crucial decision when considering how a district's budget will allocate funds to attract and retain effective educators, and it's important when we consider that “Spending on teaching staff makes up the largest share of education expenditure” (OECD, 2025). In this report, we examine how school-level socioeconomic and demographic factors help predict the average teacher salary across K–12 campuses in Texas.

Using a statewide dataset that includes information on student composition, economic disadvantage, race/ethnicity percentages, household income, and school size, we treat the problem as a supervised regression task. We build predictive models to determine:

1. Which school-level characteristics are most strongly associated with higher or lower teacher salaries?

2. How accurately can these characteristics predict a campus’s average teacher salary?

Overall, we frame this as an inference task, and want to understand how these features influence teacher salary, and by how much.

To answer the research question, we develop two different models:

- An Elastic Net model, which identifies the most influential socioeconomic and demographic predictors by shrinking weaker variables to zero

- An XGBoost gradient boosting model, which captures complex nonlinear relationships and interactions among school characteristics.

We opted for these models because they both offer a high degree of interpretability. By comparing the performance and feature importance of these two models, we are able to quantify the role that factors such as the percentage of economically disadvantaged students, neighborhood household income, racial and ethnic composition, and student population size play in shaping teacher salary levels. The findings provide insight into how local demographic conditions relate to how predictive modeling can help form policy discussions on teacher pay.

Data and Feature Selection

The dataset used in this project includes approximately 12,000 Texas school campuses. Each row represents a single school and contains covariates such as the overall school rating, median household income, demographics of students, and region of the school. These are important covariates for this problem for several reasons. An overall school rating is important because it is closely tied to teaching quality, which is influenced by salary competitiveness. School rating can also be reflective of the difficulty of work, as it may be more challenging to work at a school with an overall rating. Median Household income is important because it can be an important indicator of how much money a district has to pay its teachers. Demographics of students can also be indicative of teacher salaries because they can reveal which kinds of programs are needed for the school and what kind of funding it needs, which can influence budget decisions. Finally, the region of the school is an important feature to look at because the cost of living is very different across rural and metropolitan areas, which should also mean that salary is influenced by this feature.

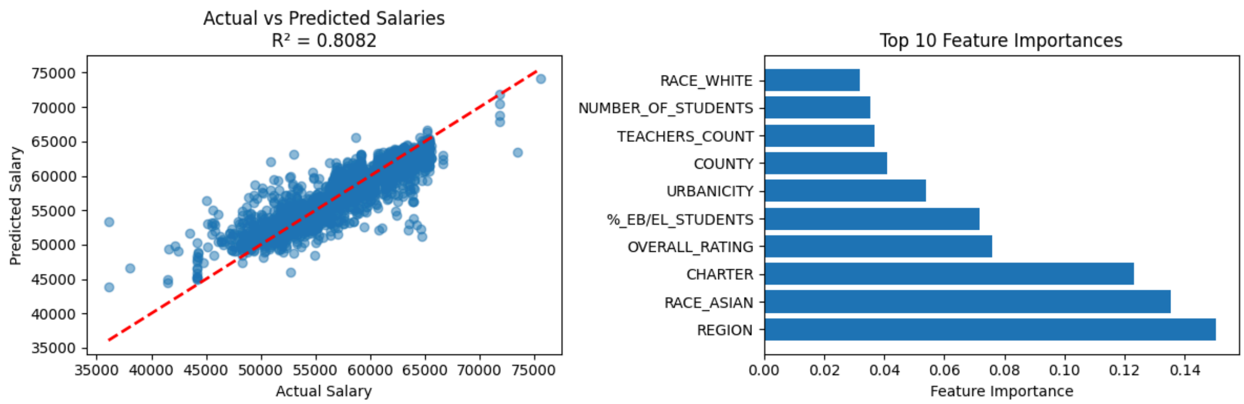
Modeling:

<u>XG Boost Model:</u>

The initial model chosen for analysis is XG Boost, which offers several advantages for this study. Its robust regularization, incorporating both L1 and L2, is crucial for preventing overfitting while maintaining the excellent balance needed to capture complex salary relationships. Furthermore, XG Boost naturally handles complex, non-linear interactions between diverse factors like demographics, location, school type, and performance metrics. The gradient-based optimization ensures robustness to outliers, effectively managing school districts with extreme salaries. A significant benefit is the clear ranking of feature importance, which aids in understanding the key drivers of salary. Finally, the model works well with mixed data types after encoding, making it suitable for our combined categorical and numerical features.

To evaluate our XG boost model, we decided to do 5-fold cross validation using a randomized 80% training, 20% testing split to maintain sufficient data for training while having a robust test set for unbiased evaluation. We use a random fold assignment rather than stratifying by region because region is included as a covariate in the model. Stratifying or blocking by region would remove between-region variation from the training process and artificially constrain the model. Under the assumption that individual observations are IID conditional on the covariates, random cross-validation provides an appropriate and unbiased estimate of out-of-sample performance.

The metrics that we are reporting are RMSE (Root Mean Squared Error), MAE (Mean Absolute Error), and the R^2 (Coefficient of Determination). For the loss function, we decided to choose squared error because, for regression tasks, it minimizes the mean squared error, providing optimal predictions under normal error assumptions. The hyperparameters that we tuned for through 5-fold cross-validation are maximum tree depth and learning rate. We found that a maximum tree depth of 6 and a learning rate of 0.1 were the best parameters.

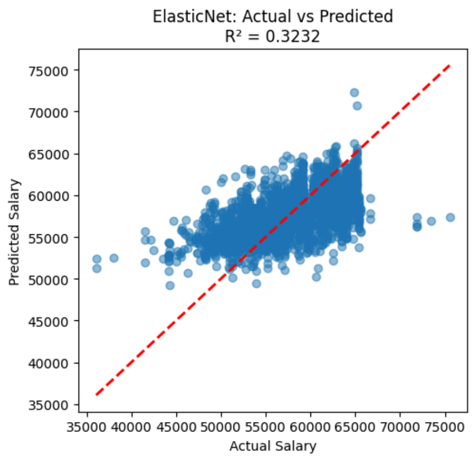


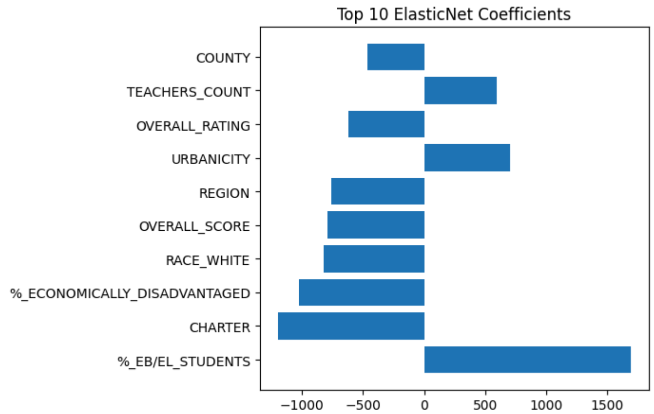
The model achieves an RMSE of $2,192.70, an MAE of $1,506.04, and an R2 of 0.8082. Thus, an XGBoost model can predict the average teacher salary within about 1.5-2.2 thousand of the observed salary at the campus level. From our XGBoost model, we found that the most important covariates in relation to teacher salary are the region, with an importance score of 0.1505, the amount of asian students, with an importance score of 0.1355, and then whether a school is a charter school or not, with an importance score of 0.1233. Therefore, when trying to plan the amount of money allocated to teachers' salaries, the most important factors are region, the composition of students, and charter status.

Elastic Net:

The second model chosen is the Elastic Net model that combines Lasso (L1) and Ridge (L2) penalties. The advantages of using this model are that it handles multicollinearity among correlated features, performs feature selection, drops less informative variables, prevents overfitting when many categorical and numeric predictors are present, and balances complexity and generalization for reliable salary predictions.

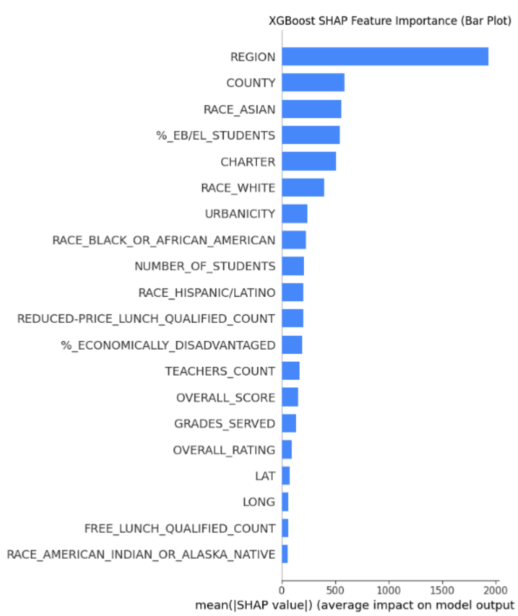
To evaluate our ElasticNet model, we use the identical cross-validation scheme to the previous model with 5-fold cross validations and 80-20 randomized split. We again decided to report RMSE (Root Mean Squared Error), MAE (Mean Absolute Error), and the R^2 (Coefficient of Determination) because these metrics provide transparent, real-world interpretability as policy and education stakeholders can directly understand “average error in dollars.” The loss function used for this model is the Mean Squared Error (MSE), as it penalizes large errors more heavily and provides optimal predictions for continuous salary values under normal error assumptions. The hyperparameters that we tuned for L1_ration, which determine the balance between lasso and ridge regularization, and the alpha, which prevents overfitting by shrinking the coefficients. The optimal hyperparameters that we found were an L1_ratio of 0.9 and an alpha of 0.1.

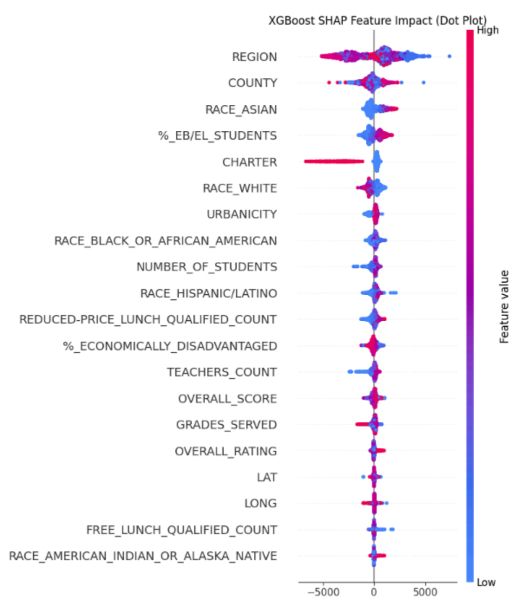




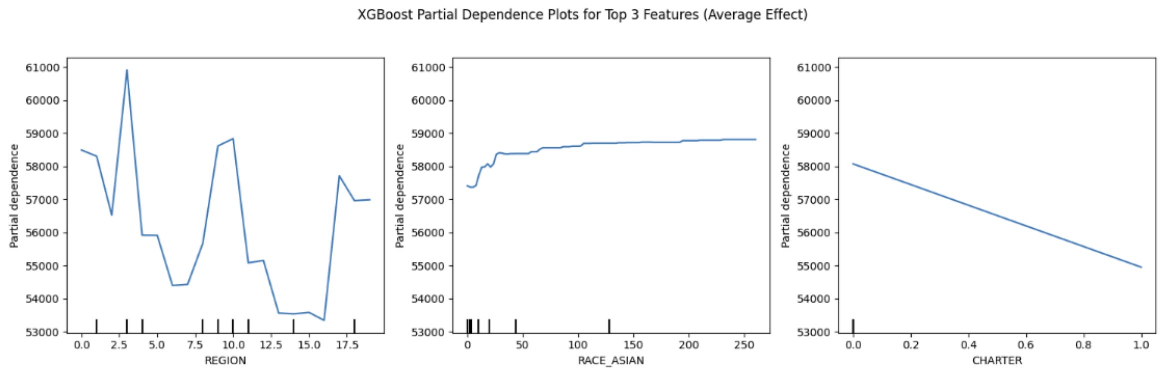
Our Elastic Net Model achieves an RMSE of $4,118.60, an MAE of $3,277.51, and an R2 of 0.3232. Thus, our Elastic Net Model can predict the average teacher salary within about 3.2 to 4.2 thousand dollars of the observed values. The most important covariates in predicting teacher salary are student composition, such as the percentage of students who are English learners (%_EB/EL_STUDENTS), the percentage of economically disadvantaged students, and whether or not the school is a charter school. From the bar graphs included, we can see that a higher percentage of English learning students means that there are higher predicted salaries, perhaps accounting for additional staffing needs. Then we see that a school being a charter school and having more economically disadvantaged students contributes to lower teacher salaries, likely meaning that charter schools rely on more donations and grant funding, and economic factors of a school's region are a heavy factor into teacher salaries. In comparison to our XGBoost model, the elastic net model is less accurate and therefore less helpful when trying to plan a teacher's salary.

Explanation:

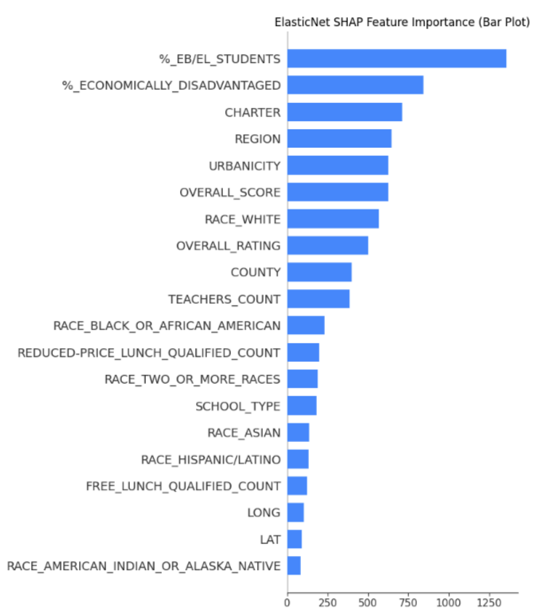


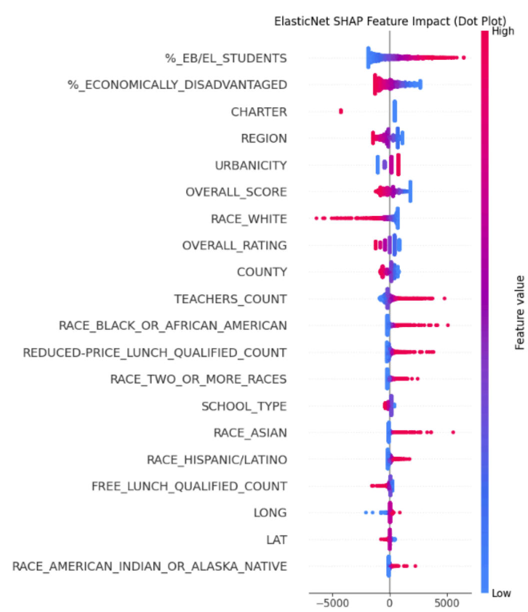


In the XGBoost SHAP Feature Importance plot, we see that REGION is easily the most important feature, exhibiting the highest average absolute SHAP value, which is then followed by COUNTY and RACE_ASIAN. These factors exert the strongest overall influence on the model features. In the SHAP Feature Impact plot we see a similar trend playover where REGION and COUNTY (Less extreme) have the widest spread of SHAP values. High values (red) push the teacher salary lower (negative SHAP value) for REGION, while for COUNTY low values (blue) push the teacher salary lower (positive SHAP value). Notably, the CHARTER shows a pronounced pattern where a high value drives the model output significantly lower.



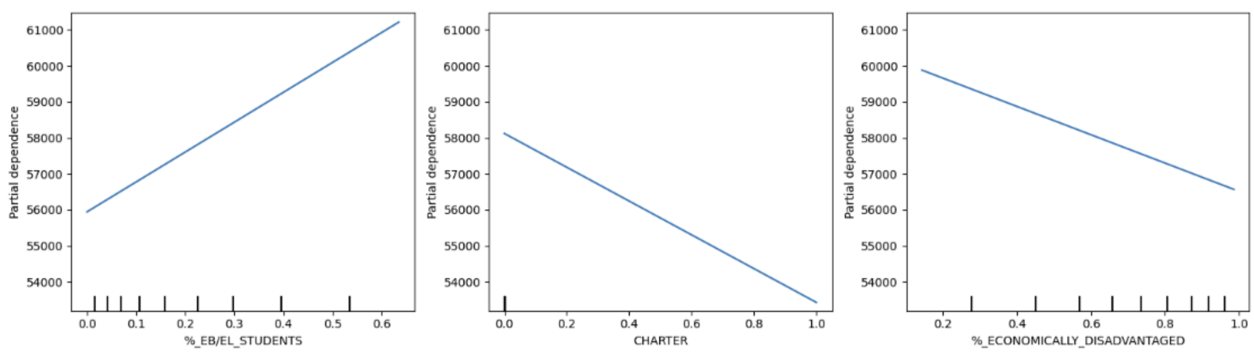
The XGBoost Partial Dependence Plots reveal distinct functional relationships for the top three features. REGION shows an extremely volatile effect, causing the predicted output to swing dramatically between approximately $61,000 and $53,500 depending on the specific region. The RACE_ASIAN feature shows the output sharply increases as the count of asians rises from zero but then plateaus after a value 50, showing no further marginal influence. Finally, CHARTER status demonstrates a clear negative linear relationship, where being a Charter school (value 1) consistently reduces the predicted output by about $3,000 compared to a non-Charter school (value 0).





In the ElasticNet SHAP Feature Importance plot, we see that %_EB/EL_STUDENTS is the feature with the greatest overall influence on the model's output, as demonstrated by the largest average SHAP value. It is followed by %_ECONOMICALLY_DISADVANTAGED, CHARTER, and a few features of a smaller but similar average SHAP value. The ElasticNet SHAP Feature Impact plot reveals a strong positive correlation for %_EB/EL_STUDENTS where high values push the teacher salary. Surprisingly we see that %_ECONOMICALLY_DISADVANTAGED has an inverse correlation where high values push the SHAP value lower. Many other features such as TEACHERS_COUNT, RACE_ASIAN, show a pronounced pattern where high values drive the model output up significantly.

ElasticNet Partial Dependence Plots for Top 3 Features (Average Effect)



The ElasticNet Partial Dependence Plots show that all three top features exhibit clear linear relationships with the predicted output. The %_EB/EL_STUDENTS feature has a strong positive effect, with the predicted output increasing consistently from approximately $56,000 to over $61,000 as the percentage rises from 0 to 0.7. Conversely, both CHARTER status and %_ECONOMICALLY_DISADVANTAGED show negative linear effects. Being a Charter school (value 1) is associated with the largest drop, decreasing the predicted output by about $4,500 compared to a non-Charter school (value 0). Similarly, an increasing percentage of economically disadvantaged students causes the predicted output to steadily decrease from about $59,500 to $56,500.

Interpretation:

The XGBoost model provided the clearest evidence about which school characteristics matter for teacher salary, achieving a very strong R^2 of 0.8082 and a low MAE of $1,506.04. This high level of accuracy means the model successfully captures real, complex patterns in the data, and its interpretations can be trusted for planning and policy discussions. In contrast, the Elastic Net model performed much worse, with an R^2 of only 0.3232 and an MAE above $3,200, showing that it fails to explain most of the variation in teacher salaries. Because of this large gap in accuracy, the findings from Elastic Net should not be relied on for interpretation. The analysis showed that several non-geographic, school-level characteristics consistently shift salaries up or down in meaningful ways, though Region and County remained the overall dominant determinants. The most influential non-geographic factor was Race_Asian, where increasing the concentration of Asian students from zero caused a sharp rise in predicted salary, plateauing around $61,000. Crucially, the model also showed that Charter schools have much lower predicted salaries, with charter status causing a clear drop of about $3,000 compared to traditional public schools

These results imply that school composition directly affects how much teachers earn, not just through broad regional factors. Because the XGBoost model is highly accurate, the magnitude of its effects likely reflects true differences in how schools are funded and allocated resources. For example, the positive effect of a higher English Learner (%_EB/EL_STUDENTS) population pushes predicted salaries upward. This is consistent with districts needing to attract teachers with specialized bilingual or ESL qualifications to meet instructional demands. This evidence challenges the "compensating differential" idea that harder-to-staff schools should pay more, as the upward pressure here appears tied to programmatic needs (e.g., specialized staffing) rather than difficult working conditions. In contrast, the simpler Elastic Net model suggested a strong negative relationship between economically disadvantaged students and salary, but its weak R^2 of 0.3232 and MAE of more than $3,200 makes its estimates less reliable, justifying the focus on XGBoost's interpretation.

Taken together, these findings reveal that school-level characteristics create real and measurable differences in campus salary structures, even after accounting for regional context. The sharp salary drop for charter schools and the distinct upward patterns for Asian and English learner concentrations indicate that staffing needs and available resources vary meaningfully across school types and student groups. Because XGBoost predicts salaries much more accurately than the Elastic Net model, its results support the conclusion that teacher pay follows predictable patterns rooted in regional economics, school type, and the composition of the student body. The consistency of these effects in a model with an R^2 over 0.80 shows that these school characteristics systematically shift salary levels, which has direct implications for policy regarding recruitment, retention, and educational equity across Texas.

Literature Discussion

In much of the current literature, teacher compensation is commonly analyzed not at the school-level but in terms of teacher-level variables, such as years of experience, certification, and individual background (Gruber 1996, Hanson et al. 2022). In addition, teacher compensation is often determined by a “salary schedule,” a standardized pay-grid used by many school districts. In a typical salary schedule, each teacher’s pay depends primarily on two factors: the number of years they have taught, and their level of education or credentials (Nittler 2019). Because of this structure, studies have determined that districts that offer a shorter progression to the top of the salary schedule effectively provide higher lifetime earnings potential, making them more attractive to early-career teachers (National Council on Teacher Quality, 2020). Other research finds that district fiscal capacity, for example, districts with higher taxable property value per student and higher maintenance-and-operations tax rates, generally have more revenue capacity, which supports higher pay (Urban Institute, 2018). Accordingly, studies also find that competition is also a significant factor to teacher salary, as if neighboring districts have better salaries, this would lead a district to pay its teachers more (Gruber 1996).

However, a key gap in the literature is how school-level and student-level factors are associated with teacher pay. Our study contributes to filling this gap by examining features such as overall school rating, percentage of economically disadvantaged students, race, and percentage of ESL students. In doing so, we find campuses with a lower school performance and higher shares of economically disadvantaged students tend to offer lower average teacher salaries. Interestingly, we also find a positive association between the share of ESL students and teacher pay, which we hypothesize may reflect districts offering higher pay to attract more bilingual or ESL-qualifier teachers. These findings contrast what a compensating-differential hypothesis might predict, i.e., more challenging work demands higher pay. This is one of the hypotheses of Gruber 1996, who in their study, analyzed how student violence rates contribute to teacher salary (Gruber 1996). They conclude that higher violence rates are correlated to lower salaries, seemingly supporting our findings that schools with more difficult working conditions seem to have lower teacher salaries.

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